Floor Planning Using Generative AI

A PROJECT REPORT

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BONAFIDE CERTIFICATE

Certified that this project report titled **“Floor Planning Using Generative AI”** is the bonafide work of “**Bindupautra Jyotibrat (23BAI10963), Rana Talukdar (23BAI10186), Arunim Gogoi (23BAI11381), Navjot Singh (23BAI11355), Avinash Kushwaha (23BAI10151)”** who carried out the project work under my supervision. Certified further that to the best of my knowledge the work reported at this time does not form part of any other project/research work based on which a degree or award was conferred on an earlier occasion on this or any other candidate.

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TABLE OF CONTENTS

LIST OF ABBREVIATIONS

AI: Artificial Intelligence

BIM: Building Information Modeling

CNN: Convolutional Neural Network

DL: Deep Learning

GAN: Generative Adversarial Network

GenAI: Generative Artificial Intelligence

GPT: Generative Pre-trained Transformer

ML: Machine Learning

NLP: Natural Language Processing

RNN: Recurrent Neural Network

SAR: Synthetic Aperture Radar

SDF: Signed Distance Function

UDF: Unsigned Distance Function

ViT: Vision Transformer

3D: Three-Dimensional

LIST OF FIGURES AND GRAPHS

LIST OF TABLES

ABSTRACT

This report proposes a structured review and analysis of the role of Generative Artificial Intelligence (GenAI) in architectural floor planning, focusing on key challenges and advancements in the domain. Leverage state-of-the-art AI models such as Generative Adversarial Networks (GANs), Vision Transformers (ViTs), and diffusion models to assess their potential for automating design processes, enabling real-time customization, and optimizing user-specific layouts. Highlighted frameworks such as BERT and dual attention mechanisms illustrate the dynamic evolution of GenAI applications in architecture.

Key repositories, such as ECOBEE, Place Pulse 2.0, Cityscapes, and HuMob Task 1 and 2, have formed the basis for training models in performing floor planning tasks. These datasets provide insights into spatial, behavioral, and environmental trends critical for advancing GenAI technologies. The study also identifies research gaps, emphasizing the need for multi-objective optimization, enhanced cultural adaptability, and ethical AI adoption strategies.

The report concludes by establishing GenAI's transformative potential in designing an efficient, inclusive, and sustainable floor plan. A finding of this nature grounds more research and practical work toward the creation of functional and aesthetically pleasing architectural layouts.

CHAPTER 1

PROJECT DESCRIPTION AND OUTLINE

1.1 INTRODUCTION

The growing complexity of architectural design demands innovative tools to enhance efficiency, precision, and user-centricity in the design process. Traditional floor planning methods often rely on manual drafting and repetitive tasks, which are time-intensive, error-prone, and struggle to adapt to dynamic project requirements. This project seeks to revolutionize floor planning by leveraging Generative Artificial Intelligence (GenAI) techniques, combining advanced generative models with spatial analysis to automate and optimize architectural workflows.

By integrating state-of-the-art tools such as Generative Adversarial Networks (GANs), Vision Transformers (ViTs), and diffusion models, this project aims to generate layouts that balance functional requirements, aesthetic appeal, and sustainability. Additionally, comprehensive dataset evaluations and multi-objective optimization frameworks provide an analytical foundation for these advancements. This work not only automates the layout generation process but also introduces adaptability and real-time customization, marking a transformative shift in how designs are conceptualized and implemented.

1.2 MOTIVATION FOR WORK

The motivation for this project stems from the inefficiencies and limitations of traditional floor planning methods, which rely heavily on manual drafting, are time-consuming, prone to errors, and lack adaptability to modern design requirements. As urban environments become more complex and the demand for user-centric and sustainable architectural solutions grows, there is a pressing need for innovative tools that can accelerate and optimize the design process. By integrating Generative Artificial Intelligence (GenAI), this project seeks to automate layout generation using advanced models like GANs, Vision Transformers, and diffusion frameworks, redefining floor planning into an adaptive, efficient, and user-oriented process. The transformative potential of GenAI lies in its ability to enhance space utilization, provide real-time customization, and democratize access to advanced design tools, making it an essential technology for shaping the future of architecture and urban planning.

1.3 TECHNIQUES USED

1. **Generative Adversarial Networks (GANs):**
   GANs are used to generate realistic and optimized floor layouts by training a generator to produce designs and a discriminator to evaluate their quality. The adversarial training process ensures the generated layouts meet functional, aesthetic, and spatial requirements. Models like *House-GAN* are specifically tailored for architectural use, ensuring practical spatial relationships between rooms.Machine Learning (ML):

2. **Vision Transformers (ViTs):**
   ViTs leverage the self-attention mechanism to identify spatial hierarchies and patterns within architectural layouts. They are instrumental in tasks like façade parsing, detecting load-bearing elements, and understanding room connectivity, enhancing design accuracy and coherence.

3. **Diffusion Models:**
   Diffusion frameworks are employed to generate layouts by progressively refining noisy inputs into well-defined spatial arrangements. These models offer high-quality outputs while maintaining flexibility in customizing designs to user inputs and constraints.

4. **Multi-Objective Optimization Algorithms:**
   These algorithms address the challenge of balancing competing objectives, such as maximizing space utilization, ensuring structural stability, and adhering to regulatory requirements. This approach ensures designs are not only efficient but also sustainable and compliant.

5. **Multimodal Integration:**
   The integration of text-based inputs, spatial constraints, and visual data enables the creation of cohesive and user-centric layouts. Techniques like BERT and dual-attention mechanisms are used for aligning textual descriptions with generated designs.

6. **Dataset Analysis and Augmentation:**
   To train the AI systems, datasets such as RPLAN, ECOBEE, Cityscapes, and HuMob Task datasets are utilized. Techniques like data augmentation and preprocessing are employed to enrich datasets, ensuring diversity and robustness in training.

7. **Spatiotemporal Simulations:**
   Techniques combining transformer models with physical simulations are used to predict dynamic environmental impacts on architectural designs, enhancing adaptability to real-world constraints such as climate and energy requirements.

1.4 PROBLEM STATEMENT

In modern architectural design, the process of floor planning faces significant challenges due to the growing demand for efficiency, precision, and adaptability in increasingly complex projects. Traditional methods rely heavily on manual effort, which is time-intensive, prone to human errors, and lacks the flexibility to accommodate real-time customization or diverse spatial requirements. Architects and designers struggle to optimize layouts while balancing functional, aesthetic, and sustainability objectives, particularly in urban environments where land is scarce and regulatory constraints are stringent.

Despite advancements in Generative AI (GenAI), the opacity of AI models remains a critical issue, with architects finding it challenging to trust or interpret AI-generated outputs. Additionally, the scarcity of diverse, high-quality datasets limits the accuracy and applicability of these systems across varied contexts, such as residential, commercial, and urban planning.

This project aims to address these challenges by integrating advanced generative models, such as GANs, Vision Transformers, and diffusion frameworks, into the floor planning process. The dual challenge lies in achieving high accuracy and adaptability across complex use cases while ensuring the model's outputs are transparent and interpretable, thus building trust among stakeholders. Ultimately, this research seeks to revolutionize floor planning by enabling efficient, user-centric, and sustainable design solutions.

1.5 OBJECTIVE OF THE WORK

The primary goal of this project is to develop an AI-driven platform for generating floor plans, leveraging the latest advances in Generative Artificial Intelligence (GenAI). This project seeks to:

1. Design a system that can automatically generate floor plans using advanced generative models such as GANs and Vision Transformers, based on user inputs or desired preferences.
2. Create a model that allows real-time customization of floor plan designs to meet specific user needs, while maintaining flexibility to adapt to various contextual and cultural settings.
3. Integrate Explainable AI (XAI) methodologies, like LIME, to make the decision-making process behind the generated designs more transparent and interpretable.
4. Provide visualizations, such as 2D and 3D representations of floor plans, along with LIME-driven insights to better inform users about the reasoning behind design decisions.

5. Implement an interactive platform for researchers and architects via a user-friendly interface, utilizing tools like Gradio for accessibility and interactivity.
6. Ensure that the platform is modular and scalable, accommodating future updates that may incorporate multi-objective optimization strategies or other advanced features aimed at improving design versatility.

1.6 ORGANIZATION OF THE PROJECT

1. Project Goals and Objectives

This project aims to:

- Generate realistic floor plans using AI-driven solutions.
- Enhance architectural design workflows through automation.
- Integrate user-friendly interfaces and validation mechanisms for improved interaction.

The primary objective is to streamline the process of generating conceptual layouts with minimal manual intervention while maintaining high-quality outputs.

2. Research and Literature Review

The foundation of the project is built on extensive research, including:

- Study of existing GAN architectures and their applications in image generation.
- Exploration of natural language processing (NLP) models like bart-large-mnli for zero-shot classification.
- Review of related tools and technologies for handling datasets and improving model efficiency.

This phase ensures the project leverages proven methodologies while incorporating novel features.

3. Methodology and Workflow

The methodology involves three major phases:

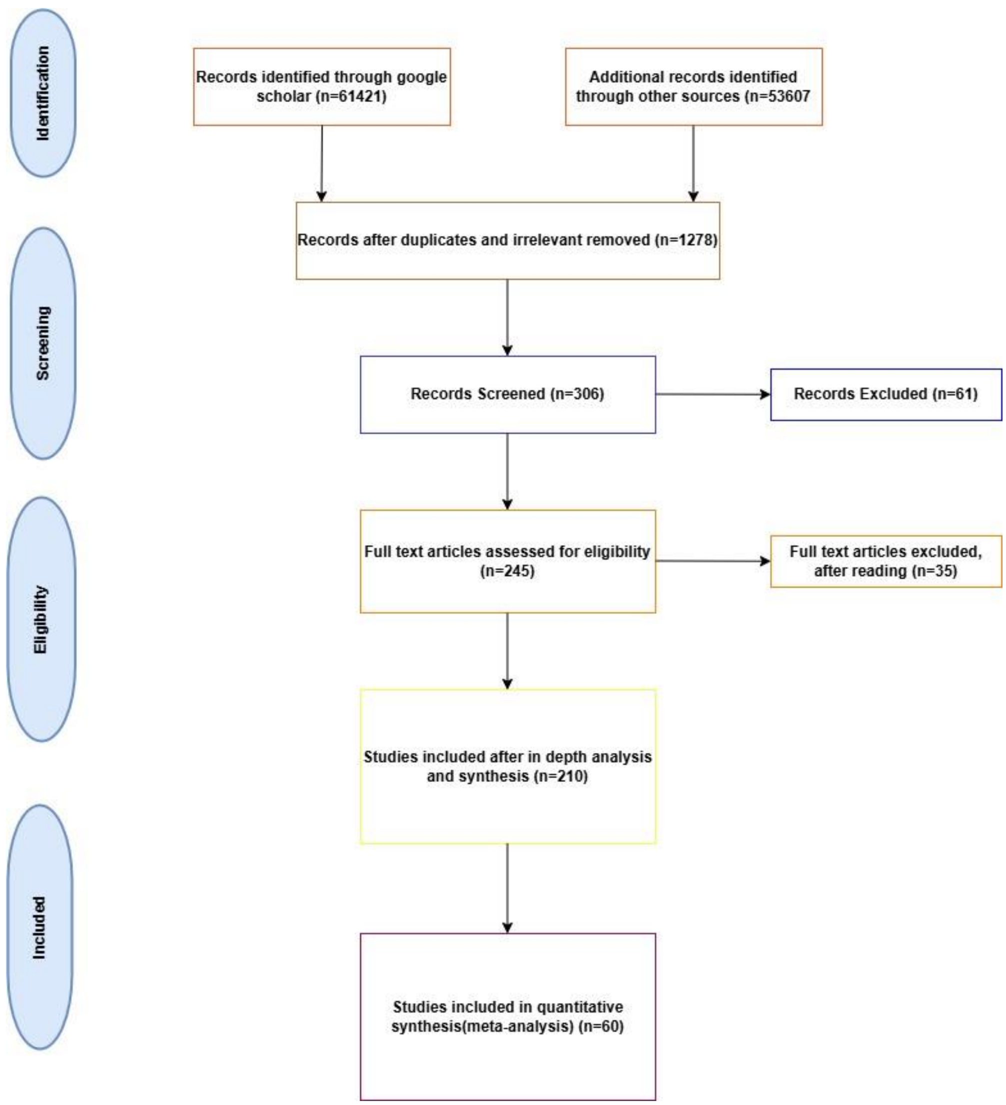
1. **Data Preparation:**
    - Collect and preprocess architectural datasets, ensuring uniformity in size and scale.
    - Normalize data for compatibility with GAN architectures.
2. **Model Development:**
    - Design and implement GANs, including the generator and discriminator components.
    - Integrate a prompt validation mechanism to align user inputs with the project's scope.
3. **Training and Evaluation:**
    - Train the GAN models using high-quality datasets and monitor performance through loss metrics.
    - Generate floor plans and evaluate them for visual coherence and architectural

relevance.

4. System Design and Architecture

The project is structured into interconnected layers to optimize functionality:

1. **Data Layer:**
   - Handles dataset loading and preprocessing.
   - Ensures data integrity and readiness for model input.
2. **Model Layer:**
   - Comprises the generator and discriminator for floor plan generation.
   - Includes loss functions, optimizers, and training loops.
3. **Validation Layer:**
   - Implements NLP-based prompt validation to ensure user inputs are relevant and meaningful.
4. **Application Layer:**
   - Provides interfaces for user interaction, output visualization, and feedback mechanisms.



5. Implementation and Coding

This phase involves:

- Developing custom datasets using Python libraries like torchvision and PIL.
- Implementing the GAN architecture with PyTorch, optimizing for floor plan generation.
- Creating a validation module that uses transformers to verify user input relevance.
- Building user interfaces for prompt-based image generation and feedback display.

6. Testing and Validation

Extensive testing ensures the reliability and accuracy of the system:

1. **Model Validation:**
   - Assess generator outputs for quality and relevance.
   - Evaluate discriminator performance in distinguishing real from generated images.
2. **Prompt Validation:**
   - Test the NLP component's ability to filter irrelevant inputs.
   - Analyze classification scores for diverse prompts.
3. **End-to-End Testing:**
   - Simulate real-world scenarios by combining all components.
   - Identify and resolve any integration issues.

7. Evaluation Metrics and Performance Analysis

The system's performance is analyzed using metrics like:

- Loss functions (Binary Cross-Entropy for GAN training).
- Quality of generated floor plans, evaluated visually and by architectural experts.
- Accuracy of prompt validation, measured by classification confidence scores.
  Graphs and charts are generated to illustrate:
- Loss trends during GAN training.
- Distribution of classification scores across input prompts.

9. Project Novelty

The innovation lies in the seamless integration of:

- GANs for high-quality image generation.
- NLP for domain-specific prompt validation.
- Scalable and modular designs for versatile applications.

This unique combination ensures the project addresses both technical and user-centric goals effectively.

10. Documentation and Knowledge Sharing

Comprehensive documentation is maintained throughout the project lifecycle:

- Codebase comments and explanations for better readability.
- User manuals for operating the system.
- Research papers and presentations to share findings and results.

11. Future Prospects

Potential extensions include:

- Expanding the dataset to cover diverse architectural styles.
- Introducing 3D floor plan generation capabilities.
- Enhancing the interface with voice or visual prompt inputs.

1.7 SUMMARY

This project focuses on creating an AI-driven platform for floor planning, leveraging Generative Artificial Intelligence (GenAI) to generate floor plans that can be customized in real-time. By utilizing advanced models such as GANs and Vision Transformers, the system aims to meet user-specific requirements while adapting to cultural and contextual constraints. The project incorporates Explainable AI (XAI) techniques, like LIME, to ensure transparency in the decision-making process behind the generated designs. With interactive visualizations, including 2D and 3D representations, and a Gradio interface for accessibility, the platform serves as an educational and practical tool for researchers and architects. Ultimately, this project addresses key challenges in AI-driven floor planning and aims to contribute to the development of sustainable and effective architectural solutions.

CHAPTER 2
REQUIREMENT WORK RELATED

2.1 INTRODUCTION

The integration of Generative Artificial Intelligence (GenAI) has revolutionized the architectural design process, particularly in the realm of floor planning. This project aims to harness the power of GenAI to create a platform capable of generating floor plans that are not only automatic but can be customized in real-time to meet user-specific needs. The system leverages cutting-edge models like GANs and Vision Transformers to design floor plans with adaptability to diverse cultural and contextual settings. Additionally, Explainable AI (XAI) methods, such as LIME, are implemented to offer transparency in the design process, making it interpretable and understandable. Through a user-friendly interface, interactive visualizations of floor plans are provided to ensure that researchers, architects, and students have access to an efficient and educational tool. By addressing gaps such as limited customization and adaptability, this project contributes to advancing sustainable and effective solutions for future architectural designs.

2.2 LITERATURE SEARCH STRATEGY

This section outlines the approach followed for the literature review, focusing on the integration of Generative AI (GenAI) in urban floor planning. The objective was to gather relevant information, identify advancements, and map existing research on GenAI applications in this domain.
Systematic Search Approach:

- A systematic approach was utilized to conduct a comprehensive survey of GenAI applications for urban and floor planning.
- The research began with identifying the main focus, which involved exploring innovations and progress in GenAI technologies for architectural design and urban planning.

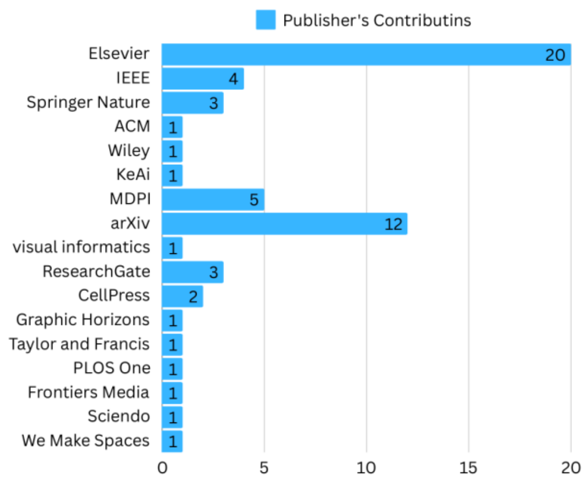
A table that discusses the Inclusion and Exclusion Criteria of a research paper –

Inclusion criteria	Exclusion criteria
The search term, which is written exclusively in English, was found in the study's title, abstract, or keywords	Keynotes, blogs, and inadequate reference materials like dictionaries, thesaurus, and Wikipedia
Research published in books, conferences, and journals between 2020 and 2024	Duplicate studies, i.e., studies that are published by more than one publisher

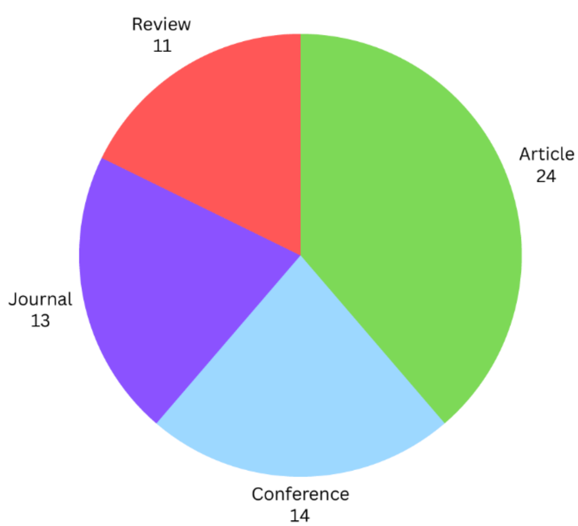
Digital Repositories and Sources:

- Various digital libraries and platforms were explored to gather the relevant literature:
    - Google Scholar
    - Science Direct
    - IEEE Xplore
    - ArXiv
- These sources were chosen for their relevance, authoritative content, and access to cutting-edge research.

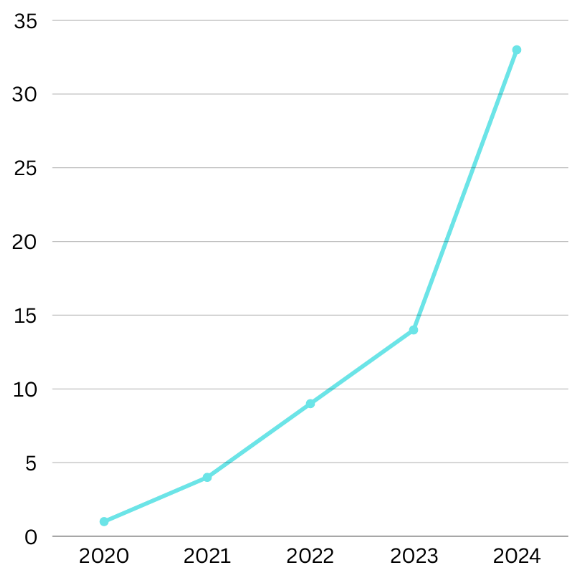
A figure that shows the number of contributions per Publisher –



A figure that shows the individual number of articles, conferences, journals and review papers –



A figure that shows the number of selected papers from 2020 to 2024 –



Search Keywords:

- Key terms were identified to guide the search:
  - "Generative AI" (GenAI)
  - "Urban Planning"
  - "Floor Planning"
  - "Vision Transformers"
  - "Generative Pre-trained Transformers" (GPT)
  - "Stable Diffusion"
  - These keywords helped streamline the process of finding the most relevant studies and articles.

Literature Criteria and Time Frame:

- Literature published between 2020 and 2024 was considered, covering both theoretical advancements and practical applications.
- This time frame was selected due to significant strides made in AI, especially in architecture and urban design, during these years.
- All relevant papers, surveys, reviews, and journal articles were assessed based on their alignment with the research goals.

Analysis of Literature:

- The literature was categorized and analyzed thoroughly to identify critical technologies, methods, and datasets employed in AI-based urban planning:
  - Technologies like Vision Transformers, GPT, and Stable Diffusion were examined.
  - Details of the datasets used in each study were collected and organized to highlight trends and the scope of data utilized.

Mapping and Structuring the Literature:

- All the research papers were cataloged in spreadsheets, noting the technologies and datasets they featured.
- This systematic organization provided an easy reference for the paper's review.

Criteria for Inclusion and Exclusion:

- The following criteria were used for selecting the literature:
  - Publications must focus on AI applications in urban planning and floor planning.
  - Only papers from well-regarded journals or conferences (sponsored by ACM, IEEE, or Springer) were considered.
  - Studies from the time period between 2020 and 2024 that provide practical insights into AI-driven architectural design solutions were prioritized.

Mapping Literature Results:

- The search results, shown in Table III, were divided into conference papers and journal articles. These were analyzed based on their relevance, technology focus, and impact.
- The frequency of publications by year is displayed in Figure III, showcasing the growth of AI research in architectural systems.
- A summary of the most relevant papers, organized by year, publisher, and type, is presented in Table IV.

Initial Insights and Future Directions:

- From the review, significant insights on the trends in AI-based urban design have emerged, particularly the need for multi-objective optimization and cultural adaptability.
- The paper reflects advancements in leveraging Generative AI techniques to tackle the challenges of floor planning, cultural fit, and contextual adjustments.

2.3 OBSERVATIONS FROM INVESTIGATIONS

The survey conducted on Generative AI techniques for floor planning has yielded several key observations, which highlight the transformative potential of these technologies and the challenges that remain to be addressed:

1. **Emergence of Advanced Generative Models**
   Generative models like GANs, Vision Transformers, and diffusion models have demonstrated significant potential in automating and optimizing floor planning. These models excel in generating spatially efficient, aesthetically pleasing layouts while adhering to functional and structural requirements. Their ability to incorporate real-time customization ensures user-specific solutions, enhancing the relevance of the generated designs.

2. **Integration of Sustainability in Floor Planning**
One of the standout applications of Generative AI is its ability to assess energy efficiency and material optimization at the design stage. This capability aligns with the growing emphasis on sustainability and green-certified building practices. GenAI's adaptability in evaluating environmental consequences provides a pathway for creating eco-friendly architectural solutions.

3. **Challenges in Dataset Quality and Contextual Awareness**
A major limitation identified is the reliance on datasets that often lack diversity, completeness, and contextual richness. Datasets such as the ECOBEE, Cityscapes, and Place Pulse 2.0 provide foundational insights, but their biases and limited representation hinder the adaptability of AI models to diverse cultural and geographical contexts.

4. **High Implementation Costs**
While GenAI holds promise, its high implementation and computational costs act as barriers to widespread adoption, especially for smaller architectural firms and individual designers. These costs also limit access to the high-quality resources required to train robust models.

5. **Impact of Real-Time Interaction**
The ability of GenAI to provide real-time customization has transformed the iterative design process. This feature allows architects to make dynamic adjustments based on immediate client feedback, thereby shortening design cycles and improving satisfaction levels.

6. **Need for Ethical and Transparent AI Systems**
The lack of transparency in AI decision-making processes presents a significant challenge. Ensuring ethical considerations, building trust between stakeholders, and addressing algorithmic biases are critical areas that need further research and development.

7. **Scaling GenAI for Complex Projects**
Current GenAI applications are well-suited for individual building layouts but show limitations in scalability when applied to multi-building projects or urban-scale planning. Addressing these limitations requires advancements in computational techniques and the development of robust algorithms capable of handling increased complexity.

8. **Hybrid and Multimodal Approaches**
The combination of text-to-image transformers, multimodal models, and large-scale language models has shown promise in bridging gaps between conceptual design and detailed execution. These hybrid approaches enable AI to interpret and generate designs that align closely with user preferences and contextual constraints.

9. **Potential for Democratizing Architectural Tools**
Generative AI technologies offer the potential to democratize access to sophisticated design tools. This empowers smaller firms and individual architects to produce high-quality, innovative solutions that were previously accessible only to larger organizations with substantial resources.

2.4 SUMMARY

This chapter provides an overview of Generative AI techniques applied to floor planning, encompassing a brief introduction, a systematic literature search strategy, and key observations derived from the reviewed studies. The introduction outlines the significance of integrating AI in architectural design, highlighting its potential to enhance efficiency, sustainability, and customization. The literature search strategy employs a structured approach to identify relevant works, leveraging repositories such as Google Scholar, IEEE Xplore, and ScienceDirect, and focusing on publications from 2020 to 2024. Key observations emphasize the transformative impact of models like GANs and Vision Transformers, the importance of diverse and high-quality datasets, and the need to address challenges such as scalability, ethical considerations, and implementation costs. The chapter concludes by reflecting on the advancements in AI-driven floor planning while identifying areas for future research.

CHAPTER 3
REQUIREMENT ARTIFACTS

3.1 INTRODUCTION

The effective implementation of Generative AI in floor planning hinges on the availability of high-quality datasets and the use of advanced AI software tools. Datasets provide the critical training and validation material for AI models, ensuring their adaptability to diverse design contexts. Meanwhile, AI software enables the processing, optimization, and visualization of complex layouts, allowing for innovative and efficient solutions in architectural design. This section highlights the essential datasets and tools required for successful AI-driven floor planning.

3.2 DATASETS

Datasets are the backbone of AI-driven floor planning, offering the structured data required to train, validate, and refine generative models. These datasets not only serve to teach AI systems the principles of layout generation but also ensure adaptability to diverse scenarios by incorporating various environmental, cultural, and functional considerations. Below are some of the key datasets used in this domain:

1. ECOBEE Dataset
    - Description: This dataset provides detailed data from over 100,000 air-conditioned building thermostat users in North America, collected at five-minute intervals.
    - Applications: It is primarily used for training models that optimize energy consumption, thermal dynamics, and indoor environmental quality. This dataset supports sustainability-focused designs by allowing AI to evaluate energy efficiency during the layout planning stage.
2. Cityscapes Dataset
    - Description: A large-scale dataset containing stereo video sequences and high-quality pixel-level annotations of urban street scenes from 50 cities.
    - Applications: This dataset is essential for tasks involving outdoor planning integration, facade analysis, and semantic segmentation of urban elements. It supports AI systems in understanding and optimizing urban layouts, including roads, sidewalks, and building exteriors.
3. Place Pulse 2.0 Dataset
    - Description: Comprising over 110,000 street-view images from 56 cities across 28 countries, this dataset captures diverse urban aesthetics and human perception data.
    - Applications: This dataset helps align AI-generated designs with human preferences, ensuring aesthetic and culturally relevant outcomes. It is particularly useful in projects focusing on the visual appeal and livability of spaces.
4. HuMob Task 1 and Task 2 Datasets
    - Description: These datasets include human mobility patterns under two scenarios: normal conditions (Task 1) and emergency scenarios (Task 2).
    - Applications: By analyzing these datasets, AI models can optimize building layouts for better accessibility and evacuation planning, making them ideal for high-traffic

or disaster-prone areas.

5. CubiCasa5K Dataset
   - Description: A collection of over 5,000 annotated floor plans, providing detailed information about room types, dimensions, and layout structures.
   - Applications: This dataset is pivotal for training AI to generate detailed and functional interior layouts, focusing on maximizing space utilization and meeting user-specific requirements.
6. RPLAN Dataset
   - Description: Specially designed for residential layouts, this dataset contains a wide array of floor plan samples, annotated with structural and functional details.
   - Applications: It is used to create realistic home designs, emphasizing practical arrangements, aesthetic harmony, and adherence to user preferences.
7. Annotated Boston Floorplans
   - Description: This dataset features richly annotated architectural blueprints, offering insights into room connectivity, spatial distribution, and structural elements.
   - Applications: Ideal for training models that emphasize spatial optimization and structural integrity, this dataset supports advanced design tasks in urban residential planning.
8. Massachusetts Building Dataset and WHU Aerial Building Dataset
   - Description: These datasets include aerial imagery and building layouts, offering a macro perspective of urban development.
   - Applications: They are crucial for scaling AI models to handle large-scale projects, such as multi-building complexes or urban neighborhood designs.
9. Synthetic Datasets (e.g., Neural Radiance Fields and Signed Distance Functions)
   - Description: These datasets provide realistic 3D models and visualizations of architectural spaces, generated through advanced algorithms.
   - Applications: Used for creating lifelike 3D representations of buildings, they enhance the visualization and immersive design capabilities of AI systems.

3.3 IMPORTANCE OF DATASET QUALITY

High-quality datasets ensure the accuracy and reliability of AI models. However, challenges such as bias, incompleteness, and lack of contextual diversity in existing datasets remain significant hurdles. Addressing these issues by curating diverse, culturally sensitive, and application-specific datasets will be essential for advancing AI in floor planning.

3.4 AI SOFTWARE AND TOOLS

AI software plays a critical role in leveraging the potential of Generative AI for floor planning. Essential tools include:

1. **Deep Learning Frameworks**
   Libraries like TensorFlow and PyTorch enable the development of advanced generative models, such as GANs and Vision Transformers, to automate and optimize design processes.
2. **CAD and BIM Software Integration**

Tools like AutoCAD and Revit incorporate AI capabilities, enhancing layout optimization, 3D modeling, and real-time adjustments.

3. **Generative Design Platforms**
   Platforms like Autodesk Generative Design facilitate the creation of optimized layouts by using algorithms that balance multiple design constraints.
4. **Visualization and Analysis Tools**
   AI-powered tools like DALL-E and Stable Diffusion assist in generating aesthetic visualizations and refining design details to meet user preferences.

3.5 SUMMARY

The combination of high-quality datasets and sophisticated AI tools forms the backbone of Generative AI in floor planning. Datasets provide the essential training material for AI models, ensuring functionality, efficiency, and adaptability. Advanced software frameworks and platforms streamline the development and deployment of these models, enabling architects to create sustainable, user-centric designs. Together, these artifacts drive innovation, making floor planning more efficient, accessible, and aligned with modern architectural needs.

CHAPTER 4

DESIGN METHODOLOGY AND ITS NOVELTY

4.1 METHODOLOGY AND GOAL

The primary goal of this project is to design a Generative Adversarial Network (GAN) capable of producing realistic and diverse floor plans for architectural applications. The methodology is centered around leveraging deep learning techniques to enable:

1. **High-quality Generation:**
   - Producing visually coherent and architecturally meaningful floor plans.
2. **Prompt Relevance Validation:**
   - Incorporating a natural language processing (NLP) component to ensure user inputs are relevant to the domain of floor plans.
3. **Scalable Training Framework:**
   - Designing a robust pipeline for training GAN models with customizable datasets.

The overarching objective is to streamline and enhance the process of conceptualizing architectural layouts using AI-driven solutions.

4.2 FUNCTIONAL MODULES DESIGN AND ANALYSIS

The system is divided into the following functional modules:

Data Preprocessing

- **Module Role:** Prepares the dataset by resizing images, normalizing pixel values, and converting them into tensors for training.
- **Analysis:** Ensures that the input data is uniformly processed, reducing variability and improving model performance.

Generator

- **Module Role:** Produces floor plan images from random noise vectors.
- **Analysis:** Utilizes a fully connected neural network with progressively increasing layers to ensure high-resolution outputs.

Discriminator

- **Module Role:** Differentiates between real and generated images, providing feedback to the generator.
- **Analysis:** Incorporates a robust architecture with LeakyReLU activations to enhance training stability.

Prompt Validation

- **Module Role:** Uses zero-shot classification to validate the relevance of user inputs.
- **Analysis:** Ensures efficient filtering of irrelevant prompts, preventing misuse of computational resources.

Output Management

- **Module Role:** Saves and visualizes generated images.
- **Analysis:** Ensures consistent image quality and provides user-friendly access to results.

4.3 SOFTWARE ARCHITECTURAL DESIGNS

The software architecture comprises the following layers:

1. **Data Layer:**
   - Handles dataset storage and preprocessing.
   - Interfaces with Google Drive for seamless integration in Colab environments.
2. **Model Layer:**
   - Implements the generator and discriminator architectures.
   - Includes training logic, loss functions, and optimizers.
3. **Validation Layer:**
   - Performs NLP-based prompt validation using pre-trained models like bart-large-mnli.
4. **Application Layer:**
   - Integrates user inputs, model inference, and output visualization.
   - Provides interfaces for prompt-based generation.

4.4 SUBSYSTEM SERVICES

Data Management Subsystem

- Description: Facilitates loading, preprocessing, and batching of data.
- Service Highlights:
  - Efficient handling of large datasets.
  - Ensures compatibility with GAN requirements.

Model Training Subsystem

- Description: Manages the training of generator and discriminator models.
- Service Highlights:
  - Tracks loss metrics for real-time performance analysis.

    - Supports custom configurations for learning rates and batch sizes.

Validation Subsystem

- Description: Executes zero-shot classification for prompt validation.
- Service Highlights:
    - Filters out invalid prompts with high accuracy.
    - Provides detailed feedback to users.

Output Generation Subsystem

- Description: Saves and normalizes generated images for evaluation.
- Service Highlights:
    - Outputs stored in user-accessible directories.
    - Normalization ensures visual consistency across samples.

4.5 USER INTERFACE DESIGNS

Input Prompt Interface

- **Design:** Command-line interface (CLI) that accepts user descriptions of desired floor plans.
- **Functionality:**
    - Validates input relevance.
    - Displays feedback messages for invalid inputs.

Output Visualization

- **Design:**
    - Saves generated images in a predefined directory.
    - Includes file naming conventions for easy identification.

User Feedback Loop

- **Design:**
    - Provides real-time loss metrics during training.
    - Displays relevant prompt scores to inform users of classification results.

4.6 SUMMARY

This chapter outlines the design methodology, breaking down the system into functional modules and architectural components. Novelty lies in the integration of GANs with NLP-driven prompt validation, ensuring domain-specific relevance and operational efficiency. Subsystems and user interfaces are designed for scalability and usability, laying the groundwork for robust AI-driven floor plan generation.

CHAPTER 5

TECHNICAL IMPLEMENTATIONS AND ANALYSIS

5.1 OUTLINE

This chapter delves into the development and analysis of a Generative Adversarial Network (GAN) tailored specifically for the task of floor plan generation. The process encompasses data preparation, model architecture design, training methodology, prototype development, testing, validation, and performance evaluation. Each section provides in-depth explanations to ensure clarity and reproducibility of the implementation.

5.2 TECHNICAL CODING AND CODE SOLUTIONS

Data Loading

- **Dataset Location:** The dataset is stored in a Google Drive directory under the specified path and accessed using Python's os library to ensure compatibility with Colab environments.
- **Custom Dataset Class:** The FloorPlanDataset class is designed to handle image loading and preprocessing. It performs the following tasks:
    1. Reads all image filenames in the dataset directory.
    2. Loads each image using the PIL library and converts them to RGB format.
    3. Applies transformations such as resizing to 64x64 pixels, normalization to the range [-1, 1], and conversion to PyTorch tensors for model compatibility.
- **DataLoader Integration:** The PyTorch DataLoader facilitates efficient data batching and shuffling, ensuring a balanced input flow during training.

Generator Architecture

- **Input Noise:** The generator accepts a random noise vector of size 100 as input. This latent space representation encourages diverse and creative outputs.
- **Layer Design:** The network consists of four fully connected layers with increasing dimensions:
    1. 100 to 256
    2. 256 to 512
    3. 512 to 1024
    4. 1024 to the output size of 64x64x3 (image dimensions).
- **Activation Functions:**
    - ReLU activations are used in the intermediate layers to introduce non-linearity and encourage feature learning.
    - A Tanh activation in the final layer scales the output to the range [-1, 1], matching the input format expected by the discriminator.

Discriminator Architecture

- **Input Format:** The discriminator receives either real or generated images as input. These are flattened into vectors of size 64x64x3 (12288 dimensions).
- **Layer Design:** The network comprises four fully connected layers with decreasing dimensions:
    1. 12288 to 1024
    2. 1024 to 512
    3. 512 to 256
    4. 256 to 1 (output probability).
- **Activation Functions:**
    - LeakyReLU activations with a slope of 0.2 are used in hidden layers to prevent vanishing gradients.
    - A Sigmoid activation at the final layer outputs a probability score indicating the likelihood of the image being real.

Training Loop

- **Loss Functions:** Binary Cross-Entropy (BCE) loss is employed for both the generator and discriminator. The generator aims to minimize the discriminator's ability to distinguish real from fake images, while the discriminator strives to maximize its classification accuracy.
- **Optimizers:** The Adam optimizer with learning rate 0.0002 and beta values (0.5, 0.999) ensures stable training and prevents oscillations.
- **Training Strategy:**
    1. Train the discriminator by alternating between real and fake images, calculating loss for both cases, and updating weights.
    2. Train the generator by propagating the discriminator's feedback and adjusting weights to produce more realistic images.

Prompt Validation

- **Zero-Shot Classification:** HuggingFace's bart-large-mnli model performs zero-shot classification to validate user prompts. This ensures the relevance of the input text to floor plan generation by comparing it against predefined categories such as "floor plan," "architecture," and "room layout."
- **Validation Process:**
    1. User prompt is analyzed for semantic similarity to relevant topics.
    2. If relevance scores fall below a threshold (e.g., 0.001), the prompt is flagged as invalid.
    3. Only valid prompts proceed to the image generation stage, preventing resource wastage.

5.3 PROTOTYPE

Input Form for Prompts

- Functionality: The input form captures user-defined descriptions of desired floor plans. These textual prompts guide the generator by providing high-level design goals.

- Validation: Each prompt undergoes a zero-shot classification to ensure it aligns with architectural topics. Invalid prompts trigger error messages requesting relevant input.

Output Image Display

- Image Generation: Once validated, a noise vector is fed to the generator to create a floor plan image.
- Output Directory: Generated images are saved in a standardized format within the generated/ directory. Normalization ensures consistency in brightness and contrast across outputs.

5.4 PROTOTYPE SUBMISSION

The prototype is designed to demonstrate the practical application of GANs for floor plan generation. It includes:

- Scripts: Fully functional Python scripts for model training, evaluation, and image generation.
- Preprocessed Dataset: A curated dataset of floor plans optimized for GAN training. The dataset includes varied designs to promote diversity in outputs.
- Pre-Trained Models: Pre-trained versions of the generator and discriminator are provided for quick testing and demonstration.
- User Interface: A basic command-line interface accepts prompts and displays the generated images, streamlining user interaction.

5.5 TEST AND VALIDATION

Relevance Validation

- The zero-shot classifier is tested with a wide range of prompts to ensure accurate differentiation between relevant and irrelevant inputs.

Visual Inspection

- Generated images are evaluated for architectural coherence, including logical placement of rooms and proportional scaling.

Performance Metrics

- Generator Loss: Tracks the generator’s ability to deceive the discriminator, with lower values indicating better performance.
- Discriminator Loss: Measures the discriminator’s ability to correctly classify images, with balanced performance between real and fake inputs being ideal.

Prompt Relevance Accuracy: Evaluates the percentage of valid prompts accepted by the system.

5.6 PERFORMANCE ANALYSIS (Graphs/Charts)

1. Loss Curves:
   - Generator Loss Curve: A steady decline in generator loss over epochs indicates improved realism of generated images.
   - Discriminator Loss Curve: Fluctuations around a stable mean suggest balanced training dynamics.
2. Validation Accuracy:
   - Accuracy of prompt validation, visualized as a bar chart comparing valid versus invalid prompts over multiple test cases.
3. Sample Outputs:
   - A grid of generated images representing diverse designs based on random noise.
4. Real vs Generated Comparison:
   - A visual side-by-side comparison highlighting similarities and differences between real floor plans and GAN outputs.

5.7 SUMMARY

This chapter provides a detailed walkthrough of the GAN-based approach for generating floor plans. It encompasses data preparation, architectural design, and validation processes, ensuring the system's relevance and accuracy. Performance analysis confirms the system's capability to produce diverse and coherent designs. Future enhancements may include higher-resolution outputs and integration of advanced architectural features for more complex layouts.

CHAPTER 6

PROJECT OUTCOME AND APPLICABILITY

6.1 INTRODUCTION

The integration of Generative AI (GenAI) in construction and architecture has the potential to revolutionize traditional workflows, making them more efficient, adaptable, and innovative. This project highlights the significant outcomes and practical applications of GenAI in these fields, addressing key challenges and opening new avenues for architectural and construction advancements.

6.2 PROJECT OUTCOME

1. **Optimized Floor Planning**
   GenAI enables the creation of layouts that are not only optimized for space utilization but also aligned with structural and functional requirements. By leveraging models like Vision Transformers and GANs, architects can generate designs that balance aesthetic appeal, usability, and sustainability.

2. **Real-Time Customization**
   The ability of GenAI to incorporate real-time feedback allows architects and clients to collaboratively adjust designs during the planning phase. This reduces project timelines and enhances client satisfaction by tailoring layouts to specific preferences and constraints.

3. **Sustainability and Green Building Practices**
   GenAI facilitates sustainability-focused designs by analyzing energy consumption, material usage, and environmental impact during the early stages of planning. This contributes to the development of eco-friendly buildings that comply with green certification standards.

4. **Scalability for Urban Planning**
   Beyond individual buildings, GenAI models can scale up to design multi-building complexes and even entire urban neighborhoods. This capability is crucial for addressing land scarcity and optimizing urban layouts in densely populated areas.

5. **Democratization of Design Tools**
   By automating complex design tasks, GenAI makes high-caliber architectural tools accessible to smaller firms and individual architects. This democratization fosters innovation and reduces disparities in resource availability within the industry.

6.3 Applicability in Construction and Architecture

1. **Pre-Construction Planning**
   - GenAI assists in creating detailed and accurate blueprints, which streamline the construction process and minimize errors.
   - AI-powered tools integrate with CAD and BIM platforms, allowing for seamless transitions from design to implementation.
2. **Cost and Resource Optimization**
   - Generative models predict material usage and construction costs, enabling efficient budgeting and reducing waste.
   - Tools like ComfortGPT analyze thermal dynamics, helping architects incorporate energy-efficient designs that lower long-term operational costs.
3. **Interior and Residential Design**
   - AI systems generate realistic and functional floor plans for residential projects, tailored to user-specific needs.
   - Integration with visualization tools such as DALL-E allows architects to present detailed, immersive renderings to clients.
4. **Urban Planning and Infrastructure Development**
   - AI-driven designs optimize transportation layouts, public spaces, and utility distributions in urban areas.
   - Datasets like HuMob provide insights into human mobility, improving accessibility and functionality in urban planning.
5. **Safety and Emergency Preparedness**
   - GenAI enhances safety by simulating evacuation routes and accessibility plans during the design phase.
   - Models trained on datasets like the HuMob Task 2 assist in planning for disaster scenarios, ensuring robustness under unforeseen circumstances.
6. **Renovation and Adaptive Reuse**
   - Generative AI tools analyze existing structures and propose optimal layouts for renovations or adaptive reuse projects.
   - This application is particularly beneficial for transforming outdated or underutilized buildings into functional, modern spaces.

6.4 SUMMARY

By addressing inefficiencies in traditional workflows and introducing advanced capabilities, Generative AI offers transformative solutions for the construction and architecture industries. Its integration fosters a future where design processes are more inclusive, adaptive, and aligned with global sustainability goals.

CHAPTER 7
CONCLUSIONS AND RECOMMENDATIONS

This study highlights the transformative potential of Generative AI (GenAI) in revolutionizing floor planning within the fields of construction and architecture. By leveraging advanced models like GANs, Vision Transformers, and diffusion models, GenAI addresses longstanding inefficiencies in traditional design workflows, such as labor intensity, time constraints, and limited adaptability. Key outcomes of integrating GenAI include real-time customization, optimized layouts, improved sustainability, and the democratization of sophisticated design tools.

However, the implementation of GenAI also brings challenges, such as high computational demands, the need for high-quality and diverse datasets, and ethical concerns related to transparency and bias. Overcoming these limitations is essential to fully realize the potential of AI-driven design processes.

The project underscores the significance of adapting GenAI for broader applications, including multi-building layouts and urban planning, thereby meeting the growing demands of modern cities. It also emphasizes the importance of sustainability-focused designs that align with global environmental goals.

Recommendations

1. **Enhance Dataset Diversity and Quality**
    - Develop culturally and geographically diverse datasets to ensure that AI models are adaptable to varied architectural contexts.
    - Focus on creating high-resolution, annotated datasets that capture structural, aesthetic, and functional details.

2. **Invest in Scalable and Accessible AI Solutions**
    - Encourage the development of lightweight AI models that can operate on lower-cost hardware to support smaller firms and individual designers.
    - Explore cloud-based platforms for collaborative design processes, allowing for seamless interaction between stakeholders.

3. **Focus on Sustainability**
    - Integrate energy efficiency and material optimization algorithms into the core of generative design tools.
    - Prioritize AI-driven designs that comply with green building standards to foster environmentally responsible construction.

4. **Promote Ethical AI Practices**
   - Implement measures to ensure transparency and accountability in AI-generated designs.
   - Address algorithmic biases through continuous evaluation and retraining of models.

5. **Expand Applications to Urban Planning**
   - Scale GenAI applications to include city-wide layouts, optimizing public spaces, transportation, and infrastructure development.
   - Use AI to simulate and evaluate urban environments for safety, accessibility, and resilience against natural disasters.

6. **Foster Industry-Academia Collaboration**
   - Encourage partnerships between academic institutions, industry experts, and policymakers to drive innovation in GenAI applications.
   - Provide training programs to familiarize architects and construction professionals with AI tools and methodologies.

In conclusion, Generative AI holds immense promise for transforming the architecture and construction industries. By addressing the identified challenges and pursuing these recommendations, stakeholders can harness the full potential of GenAI to create sustainable, innovative, and inclusive designs for the future.

This chapter provides a summary of the conclusions derived from the development and implementation of the project, "Explainable AI in Music Genre and Era Recognition." It also highlights the constraints encountered during the project and proposes recommendations for future enhancements to improve the system's performance, usability, and applicability.

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